

AI + Blockchain for Government Audit Integrity

ZasPe

Blockchain sin floro, contratos sin corrupción

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1. Executive Summary

Digital transformation in the public sector demands tools that enhance transparency, traceability, and efficiency in document management. However, many control report review processes are still carried out manually, which increases analysis times, hinders the standardization of criteria, and limits the ability to guarantee the integrity of the generated information.

ZasPe is an intelligent agent designed to automate the analysis of audit reports and institutional control documents using artificial intelligence, natural language processing (NLP), and blockchain technologies. The system is capable of interpreting natural language documents, identifying the entity under evaluation, detecting the corresponding institutional function, extracting key findings (condition, criteria, cause, and effect), generating executive summaries, and automatically formulating recommendations.

As an additional trust mechanism, each processed report generates a SHA-256 cryptographic hash that uniquely represents the certified content. This hash is recorded on the Syscoin zkSYS Testnet, providing immutable evidence of the existence and integrity of the analysis performed. This process allows for subsequent verification that the information has not been altered since its issuance, thereby strengthening document traceability and system reliability.

ZasPe's architecture integrates several modern technologies, including n8n for automated workflow orchestration, OpenAI GPT for intelligent document analysis, Node.js and Vercel for API service deployment, Telegram as a conversational user interface, and the Syscoin zkSYS Testnet as the blockchain infrastructure for information certification.

The result is a platform that significantly reduces document processing times, minimizes manual intervention in repetitive tasks, and provides verifiable cryptographic evidence regarding the generated reports. This combination of artificial intelligence and blockchain makes ZasPe a solution aimed at strengthening transparency, information integrity, and operational efficiency in audit, internal control, and institutional oversight processes.

2. The Problem

Public institutions generate thousands of administrative reports, audit findings, and control documents every year. Reviewing these documents manually is a time-consuming process that often depends on the experience and judgment of individual analysts. As a result, organizations face challenges related to inconsistent evaluations, delayed decision-making, and limited traceability of the review process.

Many existing solutions focus solely on document management or storage, without providing intelligent analysis of the report content. Consequently, identifying the type of document, extracting key findings, evaluating the audit triad (condition, criterion, cause, and effect), and generating actionable recommendations remain tasks that require significant human effort.

Another major limitation is the lack of integrity verification. Once a report is generated, there is often no mechanism to guarantee that its content has not been modified after issuance. This reduces confidence in digital reports and complicates auditing, verification, and long-term evidence preservation.

In addition, most document analysis platforms operate as isolated systems that are not easily integrated into automated workflows. Organizations frequently rely on multiple disconnected tools for document processing, artificial intelligence, notifications, report generation, and digital certification, increasing operational complexity and reducing efficiency.

ZasPe addresses these challenges by combining artificial intelligence, workflow automation, and blockchain technology into a single platform. The system automatically analyzes public-sector documents, generates structured audit reports, produces standardized PDF evidence, and records a cryptographic fingerprint (hash) of each report on the Syscoin zkSYS Testnet blockchain. This approach improves consistency, reduces processing time, enhances transparency, and provides tamper-evident certification of every generated report.

3. The Problem

Currently, the evaluation of administrative reports and audit documents in public institutions is largely performed through manual procedures. A public official or auditor receives the document, reviews its contents, identifies findings and observations, prepares conclusions, and finally archives the report for future reference.

Although this workflow has been widely adopted, it presents several limitations. The quality of the analysis depends on the reviewer's expertise, the process is time-consuming, and there is no standardized mechanism to guarantee consistency across evaluations. Furthermore, once a report is archived, there is no cryptographic proof that its contents remain unchanged, making integrity verification difficult.

Current Process Limitations

- Manual review requires significant time and human effort.
- Different reviewers may produce inconsistent analyses.
- There is no automated extraction of audit findings and recommendations.
- Reports can be modified after generation without an immutable verification mechanism.
- Traceability and authenticity depend on institutional trust rather than cryptographic evidence.

4. Why Existing Systems Fail

Organizations commonly rely on a combination of office software, email communication, document repositories, and internal information systems to manage administrative reports. While these tools facilitate document creation and storage,

they were not designed to provide automated intelligence, cryptographic integrity, or immutable evidence.

As a result, organizations continue to face challenges related to manual processing, inconsistent analyses, and the inability to prove that a document has remained unchanged after its issuance.

Comparison of Current Approaches

System	Document Storage	AI Analysis	Workflow Automation	Integrity Verification	Immutable Evidence
Manual Review	✓	X	X	X	X
Microsoft Excel	✓	X	X	X	X
Microsoft Word	✓	X	X	X	X
Email	✓	X	X	X	X
Internal Information Systems	✓	Limited	Partial	Limited	X
ZasPe	✓	✓	✓	✓	✓

Common Limitations

Most existing solutions share the same fundamental limitations:

- They require extensive manual intervention.
- They do not automatically extract audit findings.
- They cannot generate standardized executive summaries.
- They do not calculate cryptographic fingerprints of reports.
- They provide no immutable proof that a document has not been modified after generation.
- They lack blockchain-based certification for long-term verification.

Consequently, organizations must rely primarily on institutional trust rather than cryptographic evidence to validate the authenticity and integrity of their reports.

5. Our Solution

ZasPe integrates Artificial Intelligence, workflow automation, and blockchain technology into a unified platform that transforms unstructured administrative documents into certified digital evidence.

Instead of relying on multiple disconnected tools, the platform automatically processes incoming reports, performs intelligent analysis, generates structured outputs, and permanently certifies document integrity through blockchain registration.

ZasPe Processing Pipeline

Stage	Description
Document Ingestion	Receives administrative reports in PDF format.
AI Analysis	Extracts relevant information using Large Language Models.
Structured JSON	Converts unstructured text into standardized data.
Audit Evaluation	Generates findings, recommendations and observations.
Risk Assessment	Assigns priorities and evaluates audit criteria.
Hash Generation	Produces a SHA-256 fingerprint of the certified data.
Blockchain Registration	Stores the hash on Syscoin zkSYS Testnet.
Certified PDF	Generates a final report including blockchain metadata and verification information.

By separating the document from its cryptographic fingerprint, ZasPe minimizes blockchain storage requirements while preserving document integrity and enabling future verification.

6. AI Analysis Engine

The ZasPe AI Analysis Engine is responsible for transforming unstructured administrative documents into structured audit information.

The engine leverages Open IA Large Language Model to understand the semantic content of each report, identify relevant information, and generate standardized outputs according to predefined auditing criteria.

Instead of performing simple keyword extraction, the AI interprets the document context, identifies relationships between findings, evaluates regulatory compliance, and produces actionable recommendations.

Information Generated by the AI Engine

Component	Description
Executive Summary	High-level summary of the analyzed report.

Condition	Current situation identified during the evaluation.
Criterion	Regulatory or procedural requirement used for comparison.
Effect	Potential or observed consequence of the finding.
Cause	Root factor contributing to the identified issue.
Recommendation	Proposed corrective or preventive actions.
Audit Observations	Additional comments relevant to the evaluation.
Risk Level	Overall severity assessment.
Priority Level	Classification of urgency for corrective actions.
Triage Conclusion	Final evaluation and recommended follow-up.
Alerts	Key issues requiring immediate attention.
Specific Observations	Detailed findings extracted from the report.
Blockchain Payload	Structured information used for integrity certification.

Output Format

After processing, the AI produces a standardized JSON structure that can be consumed by downstream automation workflows. This structured output enables automatic PDF generation, blockchain certification, database integration, notification services, and future analytical processing without requiring additional manual intervention.

The standardized format also ensures consistency across evaluations, regardless of document length or writing style, reducing variability commonly found in manual review processes.

7. Workflow Architecture

The ZasPe architecture is designed as an automated, event-driven document processing workflow. Each document submitted by the user passes through a series of consecutive stages that transform a PDF file into a fully analyzed, certified, and traceable report.

The goal of this architecture is to eliminate manual intervention during document processing, ensuring consistency, speed, and verifiable evidence of every analysis performed.

Workflow Components

Component	Function
Telegram Bot	Receipt of documents sent by the user.
n8n	Complete workflow orchestration.
OCR	Extraction of textual content from PDF files.
Model Open IA	Document classification and generation of technical analysis.
Code Nodes	Data normalization, payload construction, and hash calculation.
Blockchain API	Hash registration on the Syscoin zkSYS Testnet.
HTML to PDF	Preparation of the certified report.
Telegram	Automatic delivery of the final report.

Process Automation

Each stage of the workflow produces a structured output that serves as input for the next, enabling a continuous processing chain without human intervention.

This architecture also facilitates the future integration of new modules, such as specialized OCR engines, multiple artificial intelligence models, smart contracts, or new blockchain networks.

8. Blockchain Layer

One of ZasPe's core components is its blockchain certification layer.

Instead of storing entire documents on the blockchain, the system records only a cryptographic representation of the analyzed content using an SHA-256 hash.

This approach reduces storage costs, preserves the confidentiality of the original document, and maintains permanent, immutable evidence of its existence.

Registration Process

The process begins with the construction of a structured payload containing the most relevant information from the analysis generated by the artificial intelligence.

The fields included are:

- Entity evaluated

- Document type
- Detected function
- Executive summary
- Condition
- Criterion
- Effect
- Cause
- Analysis date

This payload is serialized into JSON format, and an SHA-256 hash is subsequently calculated.

The resulting hash serves as the unique digital fingerprint of the analysis.

Next, ZasPe uses a blockchain wallet to issue a transaction to the Syscoin zkSYS Testnet using the Ethers.js library.

The hash is embedded within the transaction's "Input Data" field, ensuring permanent storage on the blockchain.

Once the operation is confirmed, the network returns:

- Block number
- Transaction hash
- Confirmation status
- Link to the blockchain explorer

All this information becomes part of the certificate generated for the user.

Benefits

Feature	Benefit
SHA-256	Unique digital fingerprint of the analysis.
Blockchain	Permanent and immutable record.
Transaction Hash	Verifiable certification identifier.
Explorer	Public and independent verification.
Syscoin zkSYS Testnet	Decentralized platform for testing and development.

9. Integrity Certification

Integrity certification is the mechanism by which ZasPe demonstrates that a report has not been altered since the moment it was generated.

The system uses cryptographic hash functions to produce a unique representation of the analyzed content.

Any modification—even of a single character—generates a completely different hash, allowing alterations to be detected immediately.

Certification Workflow

The payload generated by ZasPe represents the dataset deemed relevant for the analysis.

A SHA-256 hash is calculated from this payload and subsequently recorded on the blockchain via a digitally signed transaction.

The operation yields a unique transaction identifier that is linked to the generated report.

Finally, both the cryptographic hash and the transaction details are embedded into the final PDF, enabling authenticity verification using only a blockchain explorer.

Elements of Integrity

Element	Description
Payload	Certified structured information.
SHA-256	Cryptographic fingerprint of the analysis.
Blockchain	Decentralized and immutable ledger.
Transaction Hash	Unique record evidence.
Explorer	Public verification platform.
PDF	Certified document delivered to the user.

10. PDF Certification

Once the document analysis is complete and the blockchain certification is recorded, ZasPe automatically generates a technical report in PDF format.

This document consolidates all information produced during the processing workflow and serves as final evidence of the analysis performed.

The report includes both the results obtained via artificial intelligence and the information required to verify the document's authenticity.

Certificate Content

The PDF includes the following main sections:

- General document information.
- Executive summary.
- Automatic classification.
- Generated finding.
- Condition, criteria, effect, and cause.
- Identified observations.
- Risk assessment.
- Recommendations.
- Analysis conclusion.
- Blockchain record.
- Integrity certification.

Información Blockchain Incorporada

Cada informe certificado contiene los siguientes datos de trazabilidad:

Field	Description
Record ID	Unique report identifier.
Hash SHA-256	Cryptographic fingerprint of the analysis.
Transaction Hash	Blockchain transaction identifier.
Block Number	Block where the evidence was recorded.
Estado	Transaction confirmation.
Fecha	The exact moment of the search.
Explorer	Public link for verification.

Certification Evidence

The generated report demonstrates:

- that the analysis was performed by ZasPe;
- that the content was certified via blockchain;
- that there is a cryptographic hash associated with the report;
- that the evidence can be publicly verified using the block explorer;
- and that any subsequent modification to the document can be detected by re-comparing its SHA-256 hash with the one recorded on the blockchain.

11. Security

Security is a fundamental component of the ZasPe platform. The system has been designed to ensure the integrity, authenticity, and traceability of every document processed without exposing confidential information on the blockchain.

Instead of storing complete PDF documents, ZasPe generates a unique cryptographic fingerprint using the SHA-256 hashing algorithm. Only this hash is permanently recorded on the Syscoin zkSYS Testnet blockchain, while the original document remains outside the blockchain environment. This approach protects sensitive information while still providing verifiable proof of existence and integrity.

Because SHA-256 is a one-way cryptographic function, it is computationally infeasible to reconstruct the original document from its hash. Furthermore, any modification to the document, even a single character, results in a completely different hash value, making unauthorized alterations immediately detectable.

Each blockchain transaction is timestamped and permanently stored within a confirmed block, creating immutable evidence of when the certification occurred. Since blockchain records cannot be modified or deleted after confirmation, the integrity certificate remains permanently verifiable through the public blockchain explorer.

The security architecture of ZasPe therefore relies on multiple complementary mechanisms:

- SHA-256 cryptographic hashing to generate a unique digital fingerprint.
- No storage of original PDF files on the blockchain, preserving confidentiality.
- Immutable blockchain records that cannot be altered after confirmation.
- Timestamped transactions providing verifiable proof of certification time.
- Public transaction verification through the blockchain explorer.

Together, these mechanisms provide a robust integrity certification model suitable for audit, compliance, and documentary evidence management.

12. Technology Stack

The ZasPe platform is composed of several integrated technologies that work together to automate document analysis, certification, and delivery. Each layer of the architecture was selected to provide scalability, reliability, and interoperability within the overall system.

Layer	Technology
Artificial Intelligence	Model Open IA
Workflow Automation	n8n
Backend	Node.js
API Framework	Express.js
Blockchain Network	Syscoin zkSYS Testnet
Cryptographic Hash	SHA-256
Blockchain Library	Ethers.js

PDF Generation	HTML to PDF
Deployment	Vercel
Messaging Platform	Telegram Bot API
Version Control	Git & GitHub

The modular architecture enables each component to evolve independently while maintaining seamless integration across the complete workflow.

10. Future Work

Although ZasPe already provides automated document analysis and blockchain-based integrity certification, several enhancements are planned to further expand the platform's capabilities.

Future versions will incorporate Smart Contracts to automate document certification rules and enable decentralized validation processes without requiring backend intervention.

Another planned feature is the issuance of NFT-based Certificates, allowing each certified report to possess a unique blockchain identity that can be independently verified and transferred if required.

The platform will also include QR Codes embedded within generated PDF reports. These QR codes will provide direct access to the blockchain transaction and integrity verification page, simplifying the validation process for auditors and third parties.

Support for Digital Signatures will further strengthen document authenticity by allowing authorized officials to digitally sign reports before blockchain registration.

To improve decentralized storage, future releases may integrate IPFS (InterPlanetary File System), enabling optional storage of encrypted documents while maintaining blockchain references to their immutable content.

Additional research is being conducted on Zero-Knowledge Proofs (ZKPs) to allow verification of document authenticity without revealing sensitive information, improving privacy for government and institutional environments.

Finally, the long-term roadmap includes the implementation of a Decentralized Autonomous Organization (DAO) model for governance, enabling transparent community participation in protocol evolution, validation policies, and ecosystem management.

These future developments aim to transform ZasPe from a document certification platform into a comprehensive decentralized auditing ecosystem.

10. Conclusion

ZasPe demonstrates how artificial intelligence and blockchain technology can be integrated to modernize document auditing and integrity certification. By automating the analysis of institutional reports, generating structured findings, and permanently registering cryptographic evidence on the blockchain, the platform significantly reduces manual effort while increasing transparency, traceability, and trust.

Unlike traditional document management systems, which rely on centralized storage and manual verification, ZasPe provides immutable proof that an analysis existed at a specific point in time and has not been altered since its certification. The combination of AI-driven document understanding, SHA-256 cryptographic hashing, blockchain registration, and automated PDF generation creates a complete end-to-end workflow for trustworthy digital auditing.

The platform is designed with scalability in mind, allowing future integration of smart contracts, decentralized storage, digital signatures, and advanced cryptographic verification mechanisms. As organizations increasingly require transparent and verifiable digital processes, ZasPe provides a practical foundation for the next generation of intelligent auditing systems.

By combining automation, cryptographic integrity, and decentralized verification, ZasPe contributes to a more secure, efficient, and accountable approach to document certification and public-sector auditing.